CS401 · Modern Programming Practices

EEMS

Enterprise Employee Management System

MPP - Project

TEAM

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Agenda

01

Introduction & Context

Project goal, team, scope

02

Domain Classes, Attributes, and Associations

This section explains the conceptual architecture of your system.

03

Database Tables and Relationship

This section describes how the relational database schema is mapped to the object-oriented class model.

04

Flow of the Four Tasks

Explaining about tasks

Solution Overview

A back-office system for managing people, projects, and clients

WHAT IS EEMS?

EEMS(The Employment Management System) is a Java back-office application that lets an admin organize a company's employees, departments, projects, and client relationships in one place.

What the system does:

1. Manages organizational structure
2. Maintains employee records
3. Handles project management
4. Tracks work allocation
5. Manages client relationships

Used tools:

- Postgresql
- DBeaver
- Docker
- Mermaid for drawing diagram

4
core domain entities

3
junction tables (M:M)

7
database tables

4
signature task flows

Domain Classes

Four core entities plus three junction classes

Department

An organizational unit

- departmentId
- name, location
- budget

Employee

A member of the workforce

- employeeId
- fullName, title
- hireDate, salary

Project

A piece of work

- projectId
- name, status
- startDate, endDate, budget

Client

An external sponsor

- clientId
- name, industry
- contact name / phone / email

JUNCTION (M:M)

EmployeeProject

Employee ↔ Project

employeeId, projectId, allocationPct

DepartmentProject

Department ↔ Project

departmentId, projectId

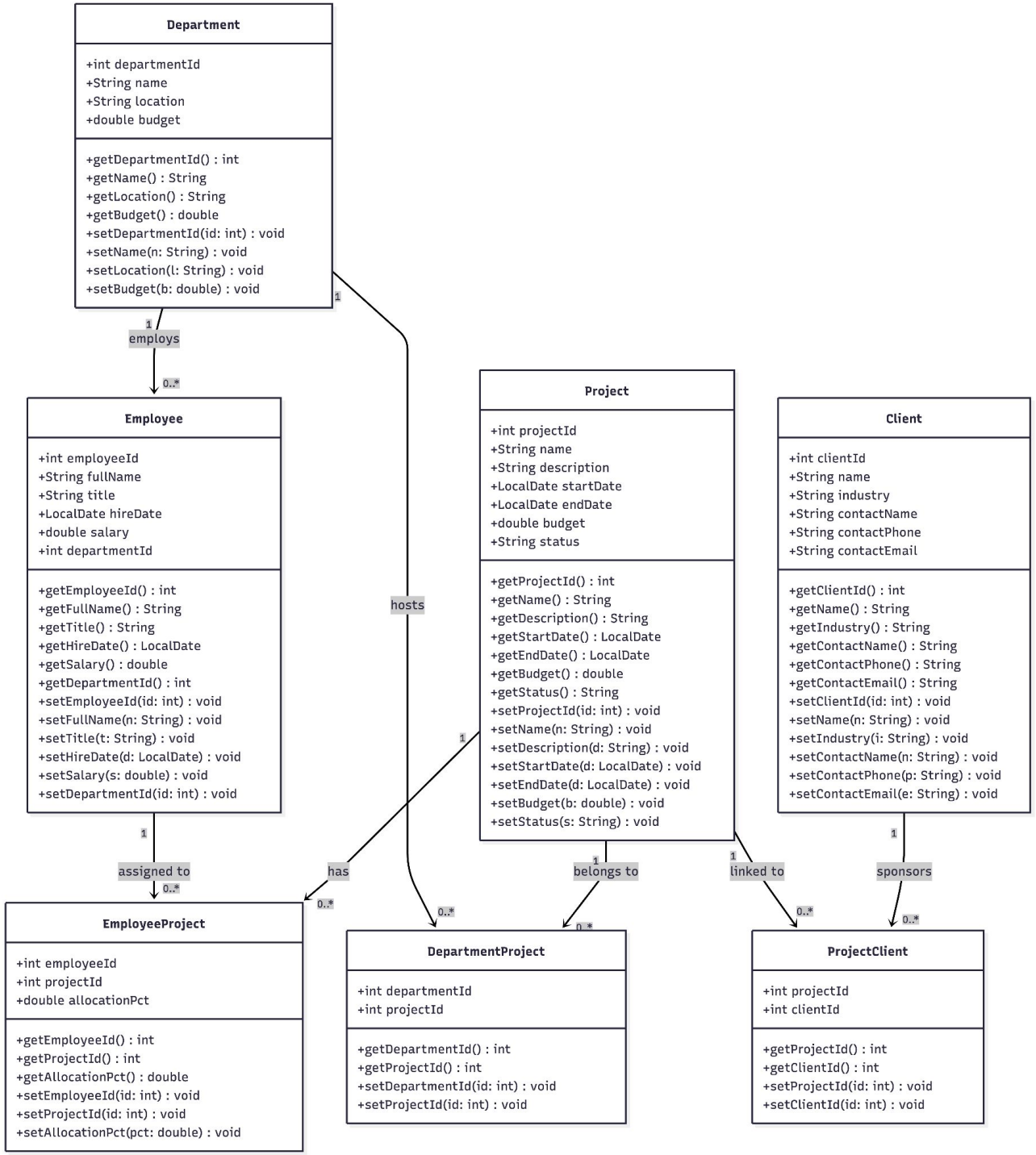
ProjectClient

Project ↔ Client

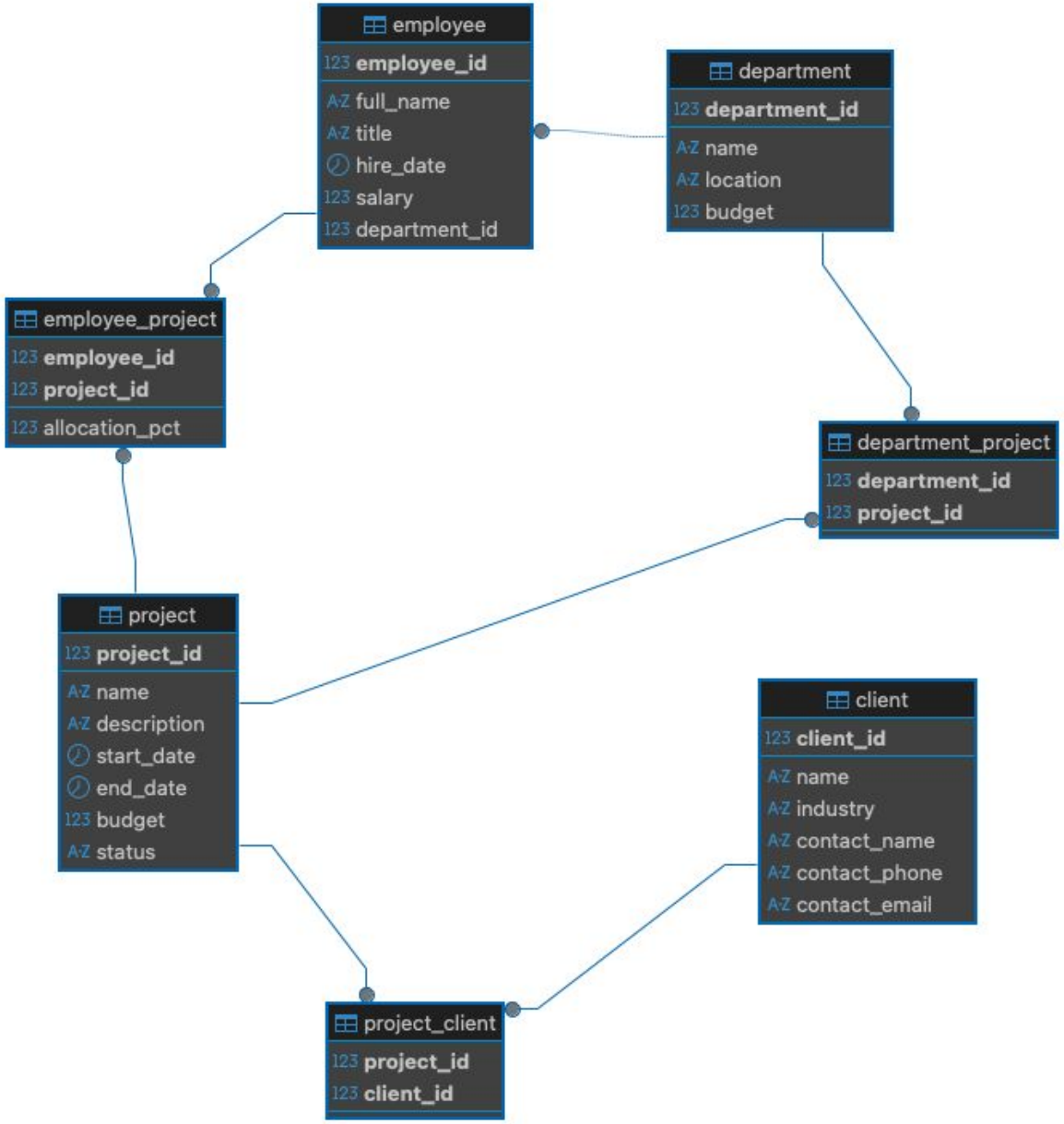
projectId, clientId

UML Class Diagram

- Department "1" --> "0..*" Employee : employs
- Department "1" --> "0..*" DepartmentProject : hosts
- Project "1" --> "0..*" DepartmentProject : belongs to
- Employee "1" --> "0..*" EmployeeProject : assigned to
- Project "1" --> "0..*" EmployeeProject : has
- Project "1" --> "0..*" ProjectClient : linked to
- Client "1" --> "0..*" ProjectClient : sponsors



Database Schema (ERD)



Thank you.